

1. What is “ Series ” ?

Series is a function used to generate a series of .geo files by changing some variables in the current .geo file using the format “From_value, to_value, step_value”. The new .geo files are saved to different project folders respectively, and all other files in the current folder will be copied to the project folders without any changes.

2. Applicable variables

Property	variable	comments
Size Info	Height	Varying device geometry to_value > from_value, from_value>0, step_value>0
	Width	
Mater	Macro Variable	Varying macro parameters to_value > from_value, from_value > 0, step_value>0
	Active Macro Variable	Varying active macro parameters to_value > from_value, from_value >0 step_value>0
Doping	Doping Concentra- tion	to_value > from_value, from_value > 0, step_value>0

3. How to change the variable in Series

The variable in Series is varying “from_value” to “to_value” with step of “step_value” . For examples:

serial - height, from 0.5 , to 1.5, step 0.6

will generate three projects with height of 0.5, 1.1, 1.5, respectively.

4. Series projects organization

After Series :

- a. A new file “series_infos.info” will be generated, which is empty and only used as monitor by other programs.
- b. A “projects” folder will be created under current folder (if there is no “projects” folder).
- c. A new folder will be created under the “projects” folder. The name of this folder should be “current project name-variable name-time”. For example, “gaas10-Height-20050419120240”. All the data of the Series will be included in this folder, we call it Series folder ;
- d. Sub-folders with the name of the variable values will be created under the Series folder. For the example above, the sub-folder names will be: 0.5, 1.1, 1.5.
- e. The .geo files with different values of the Series variable will be

copied to these sub-folders respectively. For example, .geo file with the Height of 0.5 will be copied to the 0.5 sub-folder.

- f. Copy all files (.sol, .plt, etc.) in the current project to the sub-folders.

That is it!